INTELLIGENT GROUP FORMATION AND TOPIC EVALUATION USING LLM

Project ID: 25-26J-374

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology

Specializing in Information Technology

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Declaration

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I declare that this is my own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text. Also, I hereby grant to Sri Lanka Institute of Information Technology, the nonexclusive right to reproduce and distribute my dissertation, in whole or in part in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as articles or books).

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Komila K	IT22029218	Komila

The supervisors should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

29/8/25

Signature of the supervisor

Date

(Prof. Samantha Rajapaksha)

Abstract

Forming effective student groups and selecting suitable project topics are two key factors that influence the success of undergraduate projects. However, in many universities these tasks are still handled manually or assigned randomly, often resulting in unbalanced groups, free-rider issues, and research topics that lack clarity or sufficient academic depth. The purpose of this study is to design and develop an intelligent framework that addresses these limitations by automating both group formation and topic evaluation. The research problem investigated is how to ensure fairness, diversity, and academic relevance in group projects while also improving the quality and feasibility of student project topics.

The proposed framework consists of two modules. The first is a group formation module that applies clustering algorithms such as K-Means and Hierarchical Clustering to create balanced and diverse teams based on academic performance, technical skills, and personality attributes. The second is a topic evaluation module, which uses large language models (LLMs) to analyze student-submitted project proposals and refine them for clarity, structure, and novelty. This process ensures that topics are academically sound and well-prepared for further review, while also providing transparent reasoning chains that make the refinement process explainable.

Unlike existing systems such as fuzzy clustering methods, GroupEng, or Project Zone, which focus mainly on group composition, the proposed solution combines intelligent grouping with topic refinement to enhance overall project quality. The anticipated outcome is not only fairer and more diverse student groups, but also better-defined and higher-quality project topics that can lead to stronger research outcomes. Ultimately, this study aims to improve the effectiveness of undergraduate project work and foster more meaningful collaboration in higher education.

Keywords: Group Formation, Topic Evaluation, Clustering Algorithms, Large Language Models, Educational Technology, Collaborative Learning.

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List of Abbreviations

Abbreviation	**Full Form**
AI	Artificial Intelligence
API	Application Programming Interface
LLM	Large Language Model
NLP	Natural Language Processing

DL	Deep Learning

1.Introduction

Forming effective student groups and selecting strong project topics are two of the most important factors for success in undergraduate project work. Good teams create opportunities for collaboration, knowledge sharing, and skill development, while well-defined topics provide the foundation for meaningful research. Collaborative learning has long been shown to improve both academic performance and social outcomes when students work together on shared tasks [1]. Yet, simply placing students into groups is not enough to guarantee productive teamwork. Fair allocation of members, clear assessment methods, and structured feedback are necessary to ensure effective collaboration. For instance, rubrics help provide consistent and transparent evaluations [2], while peer assessment encourages accountability and helps students develop essential soft skills [3].

In practice, however, many institutions still depend on manual or random approaches to group formation. These often produce unbalanced teams, unequal workload distribution, and weaker project outcomes. Sadeghi and Kardan [4] pointed out that random allocation, while convenient, overlooks differences in skills, academic performance, and personality, leading to poor collaboration. Similarly, Bacon, Stewart, and Anderson [5]highlighted that weak grouping methods can lower team performance and negatively affect student attitudes toward teamwork. This shows the need for systematic, data-driven approaches that consider multiple student attributes when forming groups

Researchers have started to address this problem with algorithmic methods. Zhu and Wan [6] proposed a fuzzy clustering approach where students are categorized based on prior course performance and then placed into heterogeneous groups with balanced skill levels. Dimiduk and Dimiduk [7] introduced GroupEng, an open-source tool that lets instructors define rules such as balancing GPA or clustering students with similar attributes to form fair and inclusive teams. More recently, Amarasekara et al. [8]developed Project Zone, a project management platform that integrates K-Means clustering for group formation with tools like GitHub contribution tracking and peer review, showing how machine learning can be combined with project monitoring in practice.

While these systems have advanced group formation, they largely ignore the equally important task of **topic evaluation**. Vague or poorly defined project topics often cause inefficiency, scope creep, and mismatches with supervisor expectations. In most universities, topic refinement is left entirely to supervisors, which can create bottlenecks and inconsistencies. To address this gap, researchers have begun exploring Natural Language Processing (NLP) and Large Language Models (LLMs) for evaluating project ideas. Kim [9] showed that project proposals can be converted into vector representations, allowing topics to be compared and grouped based on semantic similarity, with results validated against human evaluators. Du et al. [10] extended this idea by using LLMs to provide instant, constructive feedback on student reports, improving clarity and academic quality. More recently, Feng et al. [11] introduced GraphEval, a lightweight framework that uses LLMs and graph structures to evaluate project ideas for novelty, coherence, and logical structure. Together, these studies demonstrate the potential of LLMs to refine, assess, and strengthen student-submitted topics.

The research problem addressed in this project stems from the absence of an integrated solution that can both create balanced, diverse groups and refine project topics to ensure clarity, novelty, and academic quality. To fill this gap, the proposed framework combines clustering-based group formation with an LLM-driven topic evaluation module. Clustering ensures fair and diverse team composition, while LLMs analyze and refine student-submitted topics to make them more precise, innovative, and academically sound. This integration is expected to reduce group disparities, produce stronger project ideas, and ultimately improve the overall quality and transparency of undergraduate project management

2.Background & Literature survey

2.1 Background

Group formation and the evaluation of project topics have long been recognized as crucial elements of project-based learning and collaborative education. In earlier times, when student numbers were relatively small, forming groups or approving topics could be handled manually by instructors. Groups were often created through random allocation or by relying on a lecturer's intuition, and project topics were reviewed solely by supervisors. While these methods were manageable in smaller cohorts, the steady expansion of higher education in recent decades has revealed their limitations. Growing enrolments, increasing diversity in student backgrounds, and the demand for timely and consistent supervision have highlighted the inefficiencies of such traditional approaches [1]. Random grouping often led to unbalanced teams, with some groups excelling while others struggled, and topic evaluation carried out by supervisors alone often resulted in inconsistency and bottlenecks across departments.

During the late 20th century, some institutions attempted to introduce semi-structured methods for grouping and topic selection. For example, databases of faculty expertise or student performance records were used to guide allocation. Students were sometimes paired with peers or supervisors using keyword searches based on research interests or academic subjects. However, these keyword-based systems lacked the ability to interpret meaning at a deeper level. Small differences in phrasing could produce mismatches, leading to poor alignment between group composition and project requirements [9] At the same time, communication between students and supervisors was limited mainly to in-person meetings or email exchanges, which, while more flexible than earlier approaches, were not always effective in providing timely and continuous feedback.

The rise of e-learning platforms in the early 2000s, such as Moodle, Blackboard, and Canvas, marked the beginning of the digitalization of education. These systems provided structured environments for communication, file sharing, and course management. However, they were not designed to address the specific challenges of project-based group formation or systematic topic evaluation. They lacked intelligent allocation features and could not automatically analyze student abilities, interests, or project proposal quality [10]. As a result, administrative staff and academic supervisors still carried the burden of forming groups and reviewing topics, often without the support of advanced data-driven tools.

In the last decade, advances in machine learning and natural language processing (NLP) have offered new ways to improve both group allocation and project topic evaluation. Techniques such as clustering algorithms, semantic similarity analysis, and transformerbased architecture make it possible to analyze student performance, interests, and project descriptions with greater accuracy than ever before [12], [13], [9]. These methods support the creation of balanced, diverse groups while also helping to refine project topics for clarity, feasibility, and novelty [10], [11]. Alongside these developments, conversational agents and chatbots — especially those powered by neural networks — have become valuable in education. They provide real-time guidance, context-aware communication, and automated support, all of which are highly relevant in supervision and mentorship settings [14].

Overall, the shift from manual, intuition-based practices to AI-driven approaches reflects a broader trend in higher education toward evidence-based decision making and technological augmentation of learning processes. The background of this research project is rooted in addressing long-standing issues such as unfair or ineffective group allocation, inconsistent topic evaluation, and communication bottlenecks between students and supervisors. By leveraging clustering algorithms, NLP, and large language models, this study aims to propose a cohesive solution that ensures balanced group composition and academically rigorous topics, ultimately improving the efficiency and fairness of project-based learning in higher education.

2.2 Literature review

2.2.1 Group Formation

Over the past years, the study scenario in the Group formation and Topic Evaluation has widened, having contributions advanced by both areas of educational technology as well as computational intelligence. The three major studies are behind the scope overview in the present project as reviewed below.

GroupEng [7] is one of the early systems designed to support fairer group allocation. It allows instructors to define rules such as balancing GPA, distributing majors evenly, or clustering underrepresented students, and then prioritizing these rules based on importance. The system ensures inclusivity and fairness, though it does not extend to evaluating project topics.

Zhu and Wan [6] proposed a fuzzy clustering strategy where students are first categorized into performance levels — such as excellent, good, or average — and then combined into heterogeneous groups. This method ensures that each group has a balance of strengths and weaknesses. Their study showed that this approach improved fairness and outcomes, but it relied solely on academic performance data, overlooking other factors like interests or learning styles.

Building on these ideas, **Amarasekara et al.** [8] developed *Project Zone*, which combined K-Means clustering for group formation with features for monitoring project contributions. It included GitHub-based tracking to monitor student coding activity, peer reviews for accountability, and presentation analysis. This made it more holistic than earlier tools, but like others, it still left supervisors responsible for manually refining project topics.

More recent studies have integrated clustering with learning analytics. **Nalli et al. (2021)** [12] created a Moodle plugin that used behavioral data such as login frequency, time spent online, and assignment completion to form groups. The system first generated homogeneous clusters and then reorganized them into heterogeneous groups to maximize diversity. Their results showed a strong correlation between behavioral data and final performance, proving the effectiveness of data-informed group allocation. Similarly, a 2022 Springer study [13] emphasized the social and cognitive dimensions of group formation, demonstrating that intelligent grouping strategies can reduce the isolation of underrepresented students and foster stronger collaboration.

Holmberg (2019) developed a software tool capable of forming student groups within 60 seconds using heuristic and metaheuristic methods [15]. The study evaluated computational performance on both randomly generated and real-world datasets. Results showed that heuristic-based solutions could produce satisfactory group allocations in a very short time, making the approach promising for large-scale or timeconstrained settings. However, while efficient in execution, the system mainly focused on computational speed and optimization rather than incorporating semantic dimensions such as student interests, skills, or project topic alignment. This makes it complementary to, but narrower than, modern approaches that integrate machine learning or NLP for more context-aware group formation.

2.2.2 Topic Evaluation

While much of the research has focused on grouping, several studies have begun to address the equally important task of evaluating project topics. **Kim (2022)** [9] applied NLP methods to analyze and vectorize student project proposals. Using both wordfrequency models and contextual embeddings, the system measured topic similarity and grouped students with aligned interests. Human evaluators confirmed the accuracy of the results, but this approach did not address diversity or fairness within the groups.

Du et al. (2024) [10] explored how large language models can be used to give real-time feedback on project reports. Their system automatically generated comments and suggestions that helped students improve the clarity, structure, and academic quality of their work. This greatly reduced the workload for supervisors, though the system operated independently of group allocation.

Feng et al. (2025) [11]introduced *GraphEval*, a more advanced framework that represents project ideas as graphs, where individual arguments or concepts form nodes connected by semantic links. By applying graph neural networks (GNNs) with LLMs, GraphEval was able to evaluate novelty, coherence, and logical flow more reliably than prompt-based methods alone. While this marked a major step in topic evaluation, it was not designed to address group formation.

Dragon and Mitchell (2018) proposed *TECMap (Technology-Enhanced Concept Mapping)*, which uses teacher-defined concept maps to estimate student knowledge from assessments and group students with similar abilities [16]. The system also recommends learning resources based on shared concepts. However, TECMap's reliance on manually created domain models and its limited number of concepts (10– 40) restrict scalability and depth. Moreover, it was not evaluated against baseline grouping methods, making its effectiveness difficult to compare. Still, the study highlights the potential of concept-mapping approaches for automated group formation.

2.3 Research Gap

System / Feature	**Fair Group Allocation**	**MultiAttribute Grouping**	**Fairness / Inclusivity**	**Scalability**	**Topic Evaluation**	**Topic Novelty Detect**	**Feedback Generation**	**Integrated Group + Topic Framework**

Group Eng (2012)	✓	✗	✓	✗	✗	✗	✗	✗
Fuzzy Clustering (2019) [2]	✓	✗	✗	✗	✗	✗	✗	✗
Holmberg (2019) [4]	✓	✗	✗	✓	✗	✗	✗	✗
TEC Map (2018) [12]	✓	✗	✗	✗	✗	✗	✗	✗
Project Zone (2019) [3]	✓	✓	✗	✓	✗	✗	✗	✗
Moodle Plugin (2021) [6]	✓	✓	✗	✓	✗	✗	✗	✗

Springer Study (2022) [7]	✓	✗	✓	✗	✗	✗	✗	✗
Kim (2022) [9]	✗	✗	✗	✓	✓	✗	✗	✗
Du (2024) [10]	✗	✗	✗	✓	✓	✗	✓	✗
Feng – GraphEval (2025) [11]	✗	✗	✗	✓	✓	✓	✗	✗
Proposed System	✓	✓	✓	✓	✓	✓	✓	✓

One of the earliest attempts to support fair group allocation in classrooms was *GroupEng* by Dimiduk and Dimiduk (2012) [7]. Their system allowed instructors to set rules such as balancing GPA, distributing majors evenly, or clustering underrepresented students.

While this rule-based approach improved fairness compared to random grouping, it relied heavily on predefined heuristics and did not incorporate more adaptive methods such as machine learning or natural language processing. As a result, the system lacked flexibility and had no way to support project topic evaluation.

Building on these early efforts, Zhu and Wan (2019) [6] introduced a fuzzy clustering method where students were classified into categories such as excellent, good, or average before being reorganized into heterogeneous teams. This ensured a more balanced distribution of skills across groups. However, the model relied only on academic performance indicators and overlooked other important attributes such as personal interests or the thematic alignment of project topics, limiting its ability to fully support collaborative learning.

A more comprehensive system was developed by Amarasekara et al. (2019) [8], who designed *Project Zone*. This platform used the K-Means clustering algorithm to form groups and integrated monitoring features like GitHub-based contribution tracking, peer reviews, and presentation evaluations. By addressing accountability issues, the system added a valuable dimension beyond grouping. Yet, topic evaluation still remained entirely dependent on supervisors, leading to inconsistencies and additional workload in the project approval process.

Holmberg (2019) [15] took a different direction by focusing on computational efficiency. His system used heuristics and metaheuristics to form groups in under a minute, with successful validation on both simulated and real datasets. While the tool demonstrated scalability and speed, its focus was mainly on optimization performance. It did not account for semantic or academic considerations such as project topics, student interests, or creativity, which are equally important in project-based settings.

In contrast, some recent studies have shifted attention toward project topic evaluation rather than group formation. Kim (2022) [9] applied natural language processing (NLP) to student project proposals, using both word-frequency models and contextual embeddings to measure semantic similarity. This allowed students with related topics to be grouped together, and the approach was validated against human judgments. Although this demonstrated the value of NLP for topic alignment, the study did not address fairness or balance in group composition.

Du et al. (2024) [10] explored the potential of large language models (LLMs) to provide real-time feedback on student project reports. Their system automatically generated comments and suggestions to improve clarity, structure, and academic quality, reducing supervisors' workload. While promising, this work concentrated solely on evaluation and had no mechanism for group allocation.

More recently, Feng et al. (2025) [11] presented *GraphEval*, which used graph neural networks (GNNs) combined with LLMs to assess project ideas for novelty, coherence, and logical structure. This approach offered a transparent and structured way to evaluate academic proposals, going beyond traditional NLP models. However, like other topicfocused studies, it was limited to evaluation and did not engage with the challenges of group formation.

Taking together, these studies illustrate significant progress in both group allocation and project evaluation. Clustering approaches such as fuzzy clustering, K-Means, and heuristic/metaheuristic optimization have been useful in creating fairer groups, while NLP- and LLM-based approaches have advanced the evaluation of project topics through semantic similarity, automated feedback, and novelty detection. Yet, these two strands of research have largely developed in isolation. The key gap lies in the lack of an integrated framework that can both ensure balanced, multi-attribute group formation and carry out rigorous, automated evaluation of project topics. This research addresses that gap by proposing a unified system that brings together clustering methods for group allocation and NLP/LLM techniques for topic evaluation, offering a more complete solution for project-based learning.

2.4 Research Problem

Effective group formation and the selection of suitable project topics are two of the most influential factors in the success of academic project work. However, most institutions still rely on manual or random allocation methods to form student groups, which often leads to unbalanced teams with uneven skill distribution, free-rider issues, and reduced collaboration. Similarly, project topic evaluation is traditionally left entirely to supervisors, who manually review proposals. This creates inconsistencies, bottlenecks, and mismatches between student capabilities, project scope, and academic rigor.

Although some systems have introduced clustering algorithms, rule-based grouping, or project management platforms, they only provide partial solutions. Group allocation systems do not consider the semantic alignment of project ideas, while topic evaluation systems powered by Large Language Models (LLMs) do not integrate group formation. As a result, students face mismatches in both teamwork structure and project scope, leading to inefficiencies, weaker learning outcomes, and frustration among both students and supervisors.

The core research problem is therefore the lack of an integrated system that can simultaneously ensure balanced and fair group allocation while also evaluating and refining project topics. Without such a system, institutions continue to face inefficiencies in group formation, inconsistencies in topic evaluation, and missed opportunities to foster high-quality collaborative project outcomes.

2.5 Research Questions

- How can clustering algorithms be characterized to form balanced and diverse student groups at the early project enrollment stage, rather than relying on manual or random allocation that often produces inefficiencies?
- What methods can ensure that group formation models not only optimize fairness and diversity but also remain interpretable for instructors and administrators, so allocation decisions are trusted and accepted?
- How can NLP-driven topic evaluation systems identify vague or underdeveloped student project proposals early, while ensuring explainability and academic relevance in the refinement process?
- How can fairness-aware approaches be designed so that neither group allocation nor topic evaluation reinforce biases (e.g., favoring certain skills, genders, or popular research domains), but instead promote equitable opportunities for diverse students?

2.6 Objectives

Main Objectives

To design and develop an intelligent framework that integrates clustering-based group formation with NLP/LLM-driven topic evaluation, ensuring fair group allocation and academically rigorous project topics.

Specific Objectives

- To implement clustering algorithms (e.g., K-Means, fuzzy clustering) that form balanced and diverse student groups based on academic performance and multiattribute data.
- To develop an LLM-based evaluation module capable of refining topics for clarity, feasibility, and novelty.
- To integrate both group formation and topic evaluation into a single unified platform for project-based learning.
- To validate the system by comparing its effectiveness against traditional manual methods in terms of fairness, academic alignment, and overall student satisfactionTo evaluate the effectiveness of the system in improving supervisor–student alignment and in providing timely academic assistance through automated responses.

3. Methodology

3.1 System Overview

It is proposed that a refined mechanism for forming balanced student groups and evaluating project topics using machine learning and natural language processing should be introduced. The system is designed to address two major challenges in project-based learning: the fairness of group composition and the quality of project topics. To achieve this, the architecture integrates two complementary modules — a **K-Means clustering engine** for group formation and an **NLP/LLM-driven topic evaluation engine**.

On enrolment, student academic and skill-related information is collected, including GPA, performance in prerequisite courses, technical skills, and personal interests. This data is processed by the clustering module, which applies the K-Means algorithm to divide students into balanced and diverse groups. The algorithm ensures that no group is overpowered or underpowered by checking average group GPA against the overall cohort average, while also distributing essential skill sets fairly across groups. This guarantees heterogeneity and fairness, which are often overlooked in traditional random or GPA-only methods of group formation.

In parallel, students are required to submit their proposed project topics, typically including a title and short description. These are analyzed by the NLP/LLM module, which extracts semantic features and applies embeddings to measure similarity between topics. The system identifies overlaps or redundancies and provides instant feedback to refine clarity, feasibility, and novelty. Large Language Models (LLMs) are further employed to generate constructive suggestions for improving scope and alignment with academic standards. This ensures that each group is not only well-balanced in composition but also begins with a high-quality and well-defined project topic.

This dual architecture — combining clustering-based group allocation with intelligent, automated topic evaluation — creates an integrated environment where group formation and topic selection are handled simultaneously. On the one hand, it enhances operational efficiency and fairness in group allocation, and on the other, it strengthens project quality by ensuring that topics are semantically sound and academically rigorous. Together, these functions increase the effectiveness of project-based learning while reducing manual workload for supervisors.

System diagram

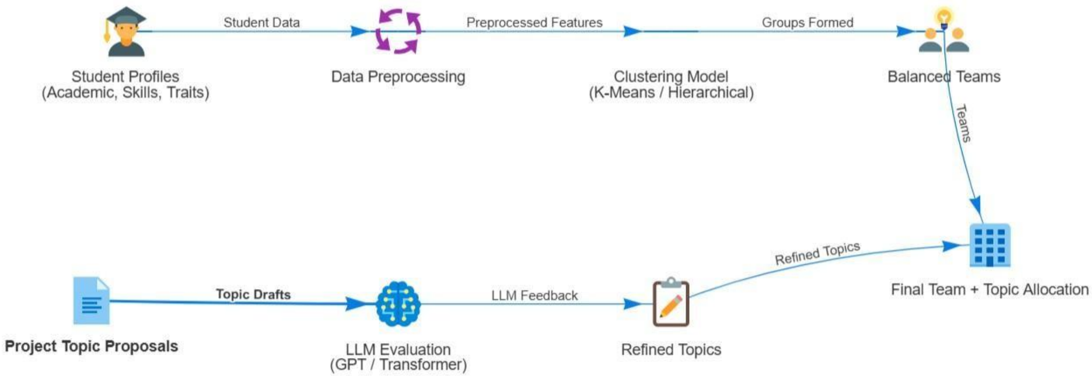

Figure 1: System Diagram

System Architecture

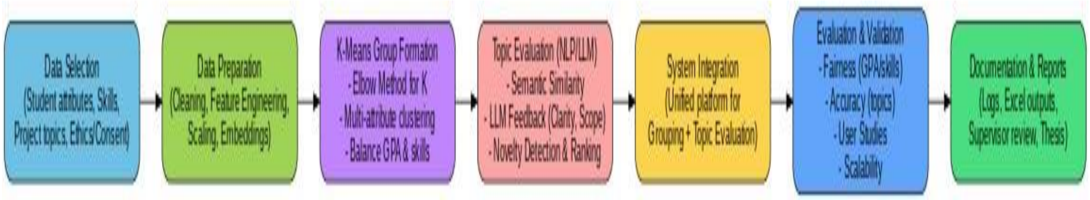

Figure 1: System Architecture

3.2 Technology

3.2.1 Frontend Development

The frontend of the system will be designed using HTML, CSS, and JavaScript, supported by a modern framework like React.js to make it responsive and easy to use. This will be the main point of interaction for both students and supervisors, accessible on laptops, tablets, or smartphones. Students will be able to fill in their academic details, add their skills, and submit project topics through simple and structured forms. Supervisors, on the other hand, will be able to view the automatically generated groups and monitor the progress of submitted topics through an intuitive dashboard.

To improve transparency, the frontend will also include visual dashboards that show group compositions, average GPA, and skill distributions. For topic evaluations, students will see visual indicators such as charts and similarity heatmaps that explain how their ideas compare with others. Notifications will also be provided in real-time so that users know immediately when groups are formed or when feedback on topics is ready. The overall design will focus on clarity, simplicity, and accessibility, making sure the system feels smooth and approachable for every user.

3.2.2 Backend Development

The backend will be built with Python Flask or Django, offering a flexible environment to connect the frontend with the machine learning modules. Think of this as the system's "control center." It will take the data submitted by students and supervisors, process it, and send it to the right modules — whether that is the K-Means clustering engine for group formation or the NLP/LLM models for topic evaluation.

The backend will also manage user authentication, role-based access, and security, ensuring that student and supervisor data are safe. It will coordinate the resubmission process when students refine their topics and will handle the delivery of automated reports and notifications. In short, the backend ties all the moving parts of the system together, ensuring that data flows smoothly and processes run reliably.

3.2.3 K-Means Clustering for Group Formation

To create fair and balanced project groups, the system will use the K-Means clustering algorithm. The algorithm takes in student data — including GPA, performance in prerequisite courses, and technical skills — and automatically groups students in a way that balances academic strengths and skill diversity. To decide how many groups should be formed, methods like the Elbow Method and Silhouette Score will be used, ensuring the chosen number of groups makes sense for the class size.

Once clustering is complete, a balancing process will be applied. This step ensures that no group is academically too strong or too weak and that skills such as programming or domain expertise are evenly distributed. This way, students are more likely to collaborate effectively and share responsibilities fairly.

3.2.4 NLP and LLM for Topic Evaluation

When students submit their project topics, the system will first clean the text using standard natural language processing steps such as tokenization, stopword removal, and lemmatization. The cleaned topics will then be converted into embeddings using models like BERT or SBERT, which understand the meaning of words and sentences. This allows the system to detect when two topics are very similar, even if they are written differently.

Beyond similarity checks, the system will also use Large Language Models (LLMs) to provide meaningful feedback on each topic. The LLM will evaluate whether the topic is clear, whether the scope is realistic, and whether the idea is novel compared to others. Students will receive constructive suggestions to refine their topics, ensuring that each group starts with a strong and well-defined project idea. A novelty scoring module will also highlight how original a topic is, reducing the chances of duplication and promoting innovative work across the class.

3.3 Implementation Strategy

3.3.1. Data Collection

The dataset for this study consists of student academic profiles, skill attributes, and submitted project topics. The academic profile includes identifiers such as student ID, GPA/CGPA, course performance records, and semester-level information. In addition to academic results, prerequisite course information is considered to better capture student progression and prepare for project-based learning.

To support balanced team formation, the dataset also incorporates demographic details (e.g., gender) and self-reported technical skills covering both programming and domainspecific areas. These attributes allow the system to distribute students fairly and ensure diversity within each group.

For the topic evaluation process, the dataset includes textual submissions of proposed project topics. Each topic contains a title and a short description, which are used for semantic similarity analysis and novelty detection. This enables the system to identify overlapping themes, refine project ideas, and align them with team competencies.

By combining academic results, prerequisite course records, skill profiles, and project topic submissions, the dataset provides a comprehensive foundation for clustering students into fair groups and for conducting automated, NLP-driven project topic evaluation.

3.3.2 Data Preparation

The dataset consists of two main components:

- Student Attributes: GPA, academic performance, technical skills, and declared interests. Unlike prior work, which considered only GPA, this system incorporates multiple features to ensure diversity and fairness in group formation.
- Project Topics: Textual project proposals submitted by students. These will be preprocessed using tokenization, stop-word removal, and vectorization to prepare them for semantic similarity and novelty evaluation.

3.3.3Group Formation using K-Means

The system applies K-Means clustering to form groups of students. The optimal number of clusters (K) will be determined using the Elbow Method based on Within-Cluster Sum of Squares (WCSS). Each student is represented as a feature vector derived from GPA, skills, and interest attributes. The algorithm proceeds iteratively by:

- Initializing centroids randomly.
- Assigning students to the closest centroid based on Euclidean distance.
- Recalculating centroids as the mean of all points in each cluster.
- Repeating until convergence or until a maximum number of iterations is reached.

Unlike traditional approaches that classify students into only *Low*, *Medium*, and *High* GPA subsets, this system extends clustering to multi-attribute grouping. Group balance is further checked by comparing the average GPA of each group against the global average, while student skills (desktop, web, or research-focused) are distributed to maximize diversity.

3.3.4 Topic Evaluation using NLP and LLMs

A unique contribution of this system is the integration of topic evaluation with group formation. Once student groups are generated, the submitted project topics are analyzed through:

- Semantic Similarity (NLP): Using word embeddings (e.g., BERT, Word2Vec), topics are compared to identify similarity, redundancy, and thematic overlap.
- LLM-based Evaluation: Large Language Models are employed to refine topic wording, provide instant feedback, and assess novelty, clarity, and academic feasibility.
- Graph-Based Evaluation (Optional): To detect novelty and logical flow, project ideas may also be represented as graphs and evaluated using graph-based techniques inspired by *GraphEval*.

This ensures that project topics are not only feasible but also original and aligned with group members' strengths.

3.3.5 System Integration

Both the K-Means grouping module and the NLP/LLM topic evaluation module will be integrated into a single platform. This integration distinguishes the system from existing solutions, which only address grouping or evaluation separately.

3.3.6 Evaluation and Validation The

system will be validated using:

- Fairness Metrics: Balance of GPA and skills across groups.

- Accuracy Metrics: Comparison of LLM topic evaluation with supervisor judgments.
- User Feedback: Surveys from students and supervisors regarding fairness, relevance, and usefulness.
- Comparative Study: The system's outputs will be compared against traditional GPAonly clustering systems to demonstrate improvement in both grouping fairness and topic quality.

3.4 Anticipated Conclusion

This research component proposes an intelligent system for student group formation and project topic evaluation, designed to overcome the limitations of traditional manual or random allocation methods. By integrating K-Means clustering for fair and balanced group formation with NLP/LLM-driven topic evaluation, the system ensures that teams are academically diverse, skill-balanced, and guided by well-defined, high-quality project topics. The anticipated outcome is a more efficient and transparent project allocation process, where students gain confidence in the fairness of grouping, supervisors benefit from reduced administrative workload, and institutions maintain higher standards of academic rigor.

3.4.1 Results

Based on insights from the reviewed literature and the proposed methodology, it is anticipated that this system will significantly improve the fairness and effectiveness of student group formation as well as the academic quality of project topics. By applying the K-Means clustering algorithm across multiple attributes such as GPA, prerequisite course performance, and technical skills, the system is expected to minimize imbalances and prevent the creation of underpowered or overpowered groups. This ensures that every student has an equitable opportunity to contribute and benefit from teamwork.

The integration of NLP and LLM-driven topic evaluation is likely to enhance the clarity, novelty, and feasibility of project ideas by providing automated, constructive feedback and detecting redundancies or overlaps across submissions. This will not only reduce the likelihood of vague or repetitive project topics but also assist students in refining their ideas to meet academic standards. As a result, supervisors will experience reduced workload in manually reviewing preliminary topics, while students will gain more confidence and direction in their project work.

Overall, the system is expected to lead to more balanced teams, higher-quality project topics, and a streamlined academic workflow. This will improve collaboration, reduce mismatches, and foster more effective project outcomes, ultimately enhancing both student and supervisor satisfaction.

3.4.2 Application

The proposed system is best suited for academic environments where large numbers of students need to be grouped for project work and their research topics evaluated efficiently. Universities and higher education institutions can greatly benefit from its ability to form fair and balanced student teams, while also ensuring that project topics are novel, feasible, and academically rigorous. The integration of K-Means clustering for group allocation and NLP/LLM-driven feedback for topic refinement ensures smoother project initiation processes, reduces mismatches, and improves overall project quality. Supervisors will also benefit, as the system minimizes the manual workload of reviewing every project topic at the preliminary stage, allowing them to focus on higherlevel mentoring and academic guidance.

3.4.3 Real-World Use

In real-world academic scenarios, the system can serve as a practical solution for managing both student grouping and project topic evaluation across a wide range of programs and disciplines. Its flexibility means it can be applied not only in computer science or engineering courses but also in interdisciplinary projects where balanced teams and clear, well-defined topics are equally critical. Beyond academia, the system has the potential to support professional training programs, internship coordination, and corporate learning environments where project-based learning is common. In such contexts, the ability to intelligently match participants into teams and refine their project ideas ensures efficiency, fairness, and better learning outcomes.

3.5 Commercialization

The proposed framework has strong potential for commercialization, extending beyond its role as a research prototype into a scalable, market-ready product. Academic institutions could adopt the platform on a subscription basis to streamline project allocation and reduce administrative costs while maintaining high standards of project supervision. In addition, professional-training organizations, corporate internship programs, and research institutes could use the platform to form balanced teams and evaluate project ideas in a structured manner. Commercialization strategies may include **cloud-based hosting**, **institutional licensing**, and **custom integration services** tailored to the needs of different organizations. By offering flexible deployment options and subscription-based pricing models, the system can generate sustainable revenue streams. Over time, this adaptability will transform the framework into a commercially viable product with long-term relevance in both education and professional training sectors4.

4.Gannt chart

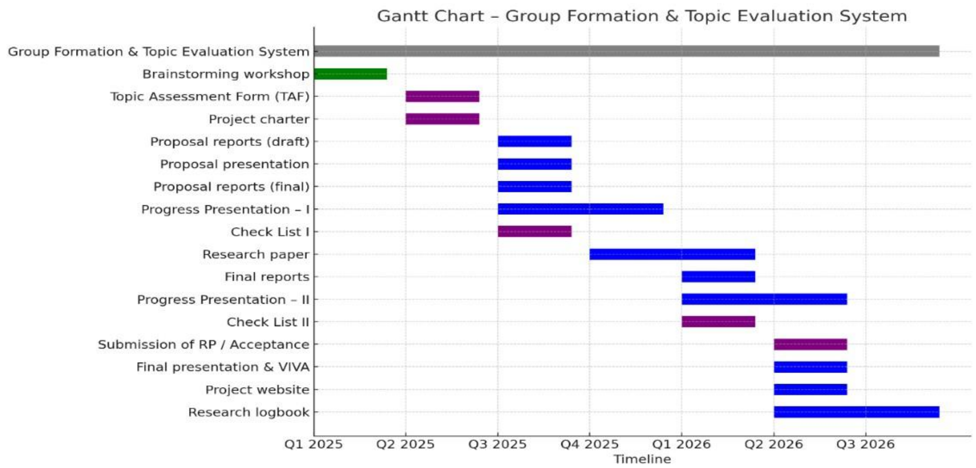

Figure 2: Gannt chart

5. Work Break Down Table

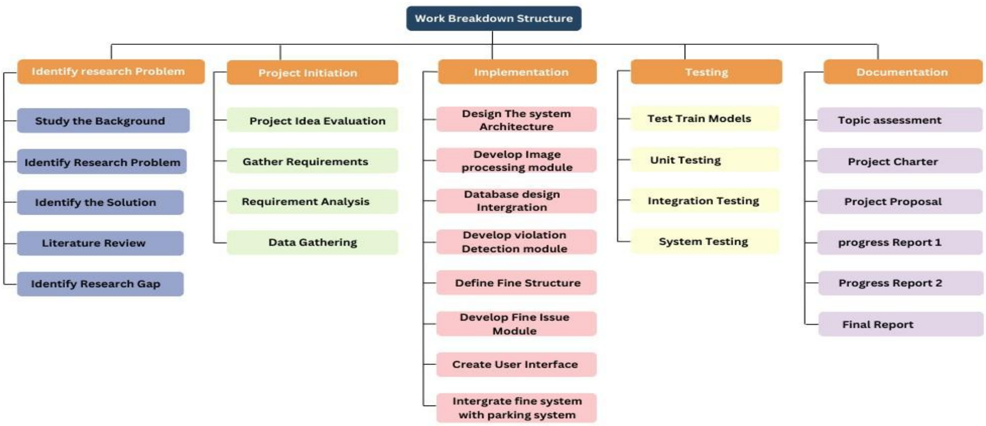

Figure 3: Work Break Down Table

6. Project Requirements

6.1 Functional Requirements

- The system will allow students to submit project titles and short descriptions.
- The system will capture academic data such as GPA, prerequisite course results, and technical skills.
- The system will apply K-Means clustering to automatically generate balanced groups.
- A post-processing module will ensure fair group composition by balancing GPA, skills, and diversity.
- The system will evaluate submitted project topics using NLP/LLM techniques for clarity, novelty, and feasibility.
- The system will detect duplicate or semantically similar topics using embeddings.

- The system will generate automated feedback on project topics to guide refinement and improvement.
- A dashboard will display group allocations, group metrics, and topic evaluation results.
- Students will be able to resubmit improved topics for re-evaluation.
- The system will generate downloadable reports of group assignments and topic evaluations.

6.2 User Requirements

- Students can enter academic and skill-related details during enrollment.
- Students can submit project topics and receive automated evaluations and feedback.
- Students can view their assigned groups and topic feedback through the portal.
- Supervisors can review group metrics and topic evaluation results.
- Supervisors can approve, reject, or request refinements on project topics.
- Administrators can upload cohort data and monitor group formation.
- All users will log in securely with unique credentials.
- The system will work across devices including desktops, tablet, and mobile.
- Students will receive notifications when groups are formed, or topic feedback is available.

6.3 System Requirements

- The frontend will be developed using HTML, CSS, and JavaScript with a modern framework.
- The backend will be implemented in Python with REST APIs.
- K-Means clustering will be applied to student data for group formation.

- NLP/LLM models will process project topics for evaluation and feedback.
- The system will run on modern browsers such as Chrome, Edge, Firefox, and Safari.
- The server will have at least 8GB RAM and a multi-core processor to handle clustering and NLP tasks.
- Topic embeddings will be stored in a vector database.
- All communications and data transfers will be encrypted and secured with HTTPS.
- The system will support large cohorts without performance degradation.
- The interface will be mobile-responsive and user-friendly.

6.4 Non-Functional Requirements

- The system should be able to identify magnesium deficiency, leaf scorch decline, and water resources with high accuracy and speed.
- Interfaces should be more user-friendly.
- The application should be reliable.
- Mobile Applications should properly work for ios and android devices.

6.5 Budget and Budget Justification

Item/Resource	**Estimated Cost (LKR / USD)**
Development Tools (IDE, Libraries, APIs)	30,000 LKR / 95 USD
Hosting & Deployment (Cloud/Server)	50,000 LKR / 160 USD
Hardware (Laptop/Workstation)	150,000 LKR / 500 USD
Internet & Communication	15,000 LKR / 50 USD

Item/Resource	Estimated Cost (LKR / USD)
Development Tools (IDE, Libraries, APIs)	30,000 LKR / 95 USD
Documentation & Reporting	12,000 LKR / 40 USD
Contingency (10%)	26,000 LKR / 85 USD

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